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Lightning – Talk

AI



Distributing AI-Inference to Multiple Machines

... for larger models or
faster speed

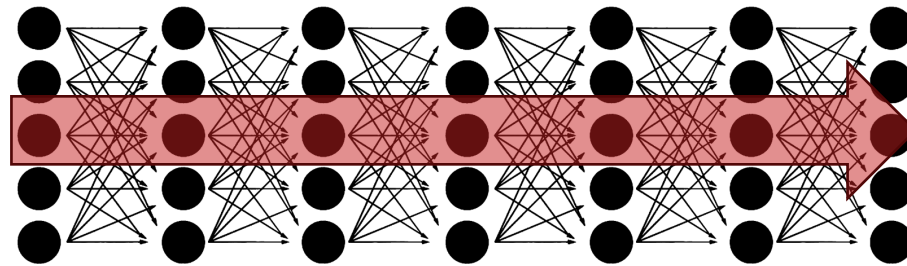
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ClubComputer.at
... begleitet Dich in die digitale Zukunft.

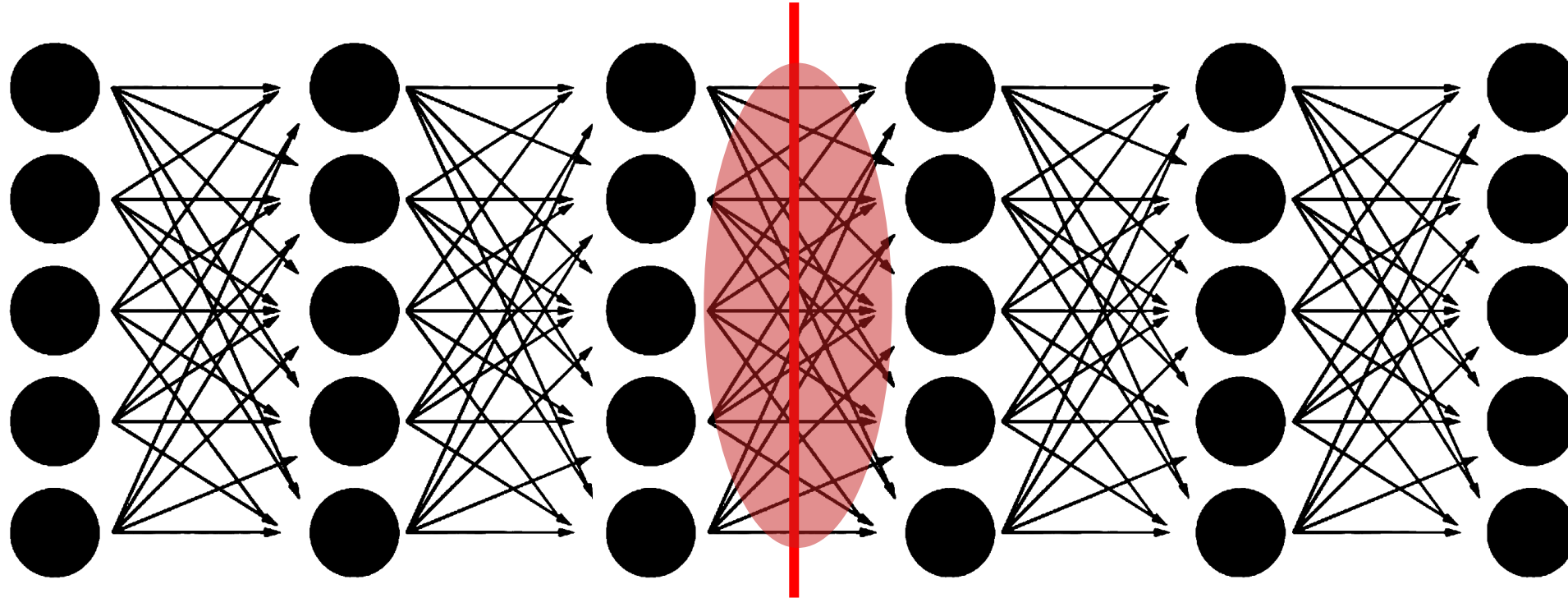
When you need MORE ...

- Your model is too large for one machine
- You want more speed
 - Prompt-processing/**prefill** is mainly **compute** bound
 - Token-generation/**decode** is mainly **memory-bandwidth** bound



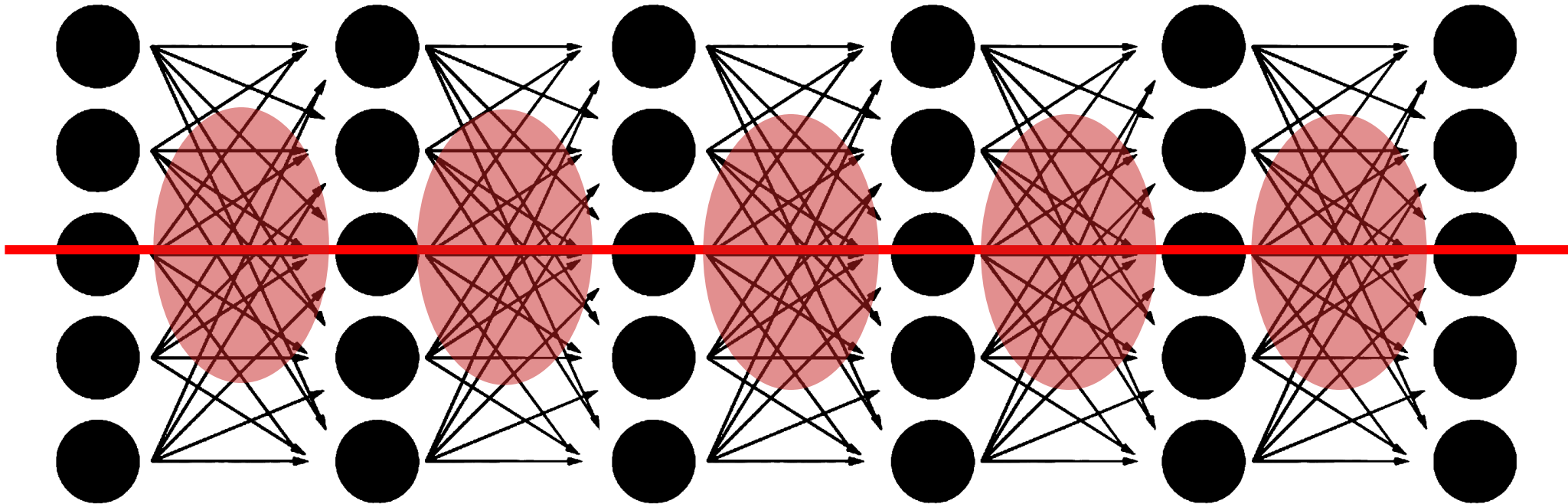
- Solution: Distribute the Artificial Neural Network (ANN) to **multiple machines**

Easier: **Pipeline P**arallelism



- Cutting ANN „vertically“ between machines – **communicating only once**
- No ultra-high bandwidth network needed, but RDMA is still needed
- No TG speed improvement, but **batching-improvements**

Easier: **T**ensor **P**arallelism



- Cutting ANN „horizontally“ between machines – ***communicating at each layer***
- **ANN head-size needs to be divisible by number of machines**
- Ultra-high bandwidth network with RDMA needed
- **TG speed improvement**, no batching-improvements



Hardware+Software Support

- Standard TCP/IP is too slow (latency is key) – you need RDMA
- Best: Thunderbolt5 (80GB/s) or NVLink (200GB/s)
- Software:
 - Most common – **vLLM** (both TP and PP, see its recipes)
 - Exo – sorry, no detailed knowledge
 - MLX on Macs supports RDMA over TB5
 - Llama.cpp experimental support for “RPC” (incl. RDMA)
distributes from “head”/RPC-backend to “workers”/ggml-rpc-servers